Thermal physics

Mark scheme and worked solutions for 60 questions: 30 multiple choice, 30 structured.

1 · Part 1

Core · States of matter · [1]

Answer: C

Worked solution

Liquid particles are close together, so there is almost no space to squash them into (not easily compressed, fixed volume). They can move past each other, so a liquid flows and has no fixed shape.

2 · Part 1

Core · Melting, boiling and evaporation · [1]

Answer: A

Explanation

While ice melts, the temperature stays at 0 °C even though energy is supplied, because the energy is used to overcome the attractive forces holding the particles in fixed positions; A is right. B and D wrongly say the temperature rises during melting. C is wrong because if the particles moved faster (gained kinetic energy) the temperature would rise.

3 · Part 1

Core · Particle model · [1]

Answer: B

Worked solution

As temperature falls, particles move more slowly and have less kinetic energy. There is a lowest temperature, −273 °C, where the particles have the least kinetic energy they can have. This is absolute zero, which is 0 K on the kelvin scale.

4 · Part 1

Core · Particle model · [1]

Answer: D

Worked solution

This is Brownian motion. Air molecules are too small to see, but they move randomly at high speed. They bombard each smoke particle from all sides. Because the number and strength of hits on each side are not exactly equal at each moment, the smoke particle is pushed first one way and then another. This is evidence that air is made of moving particles.

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5 · Part 1 Core Particle model [1]

Answer: A

Explanation

Adding air at constant volume and temperature puts more particles in the same space, so they hit the inside of the tyre more often, giving a larger pressure; A is right. B is wrong because the temperature, and so the speed of the particles, has not changed. Particles do not change size (C), and gas pressure is not caused by particles repelling each other (D).

6 · Part 1 Supplement Particle model [1]

Answer: B

Worked solution

Use the words carefully: "molecules" are the tiny particles of water; "microscopic particles" (such as pollen grains) are large enough to see under a microscope. The water molecules cannot be seen, but their random collisions with the pollen grains make the grains move randomly.

7 · Part 1 Supplement Particle model [1]

Answer: C

Worked solution

$p = F / A$

$p = 150\ \text{N} / 0.050\ \text{m}^2 = 3000\ \text{Pa}\ (3.0 \times 10^3\ \text{Pa})$.

Check: $3000\ \text{Pa} \times 0.050\ \text{m}^2 = 150\ \text{N}$, which matches the force given.

8 · Part 1 Core Gases and the absolute scale of temperature [1]

Answer: A

Worked solution

$T = \theta + 273 = -18 + 273 = 255\ \text{K}$.

Check: $255 - 273 = -18\ ^\circ\text{C}$.

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9 · Part 1 Core Thermal expansion of solids, liquids and gases [1]

Answer: C

Explanation

When a solid is heated, its particles vibrate more and, on average, move further apart, so the rod expands; C is right. The mass (A) and the number of particles (D) do not change, and the particles themselves do not get bigger (B).

10 · Part 1 Core Thermal expansion of solids, liquids and gases [1]

Answer: D

Explanation

For the same temperature rise, the metal lid expands more than the glass, so the lid becomes looser and is easier to turn; D is right. A would make the lid tighter. B and C are wrong because heating causes expansion, not contraction.

11 · Part 1 Core Specific heat capacity [1]

Answer: B

Explanation

Raising the temperature of an object increases its internal energy, while its mass stays the same, so B is right. A reverses the energy change, and C and D wrongly suggest that heating adds mass.

12 · Part 1 Supplement Specific heat capacity [1]

Answer: D

Worked solution

Temperature rise $\Delta\theta = 35 - 20 = 15\ °C$.

$c = \Delta E / (m\Delta\theta) = 12\,600 / (0.40 \times 15) = 12\,600 / 6.0 = 2100\ J / (kg\ °C)$.

Check: $\Delta E = mc\Delta\theta = 0.40 \times 2100 \times 15 = 12\,600\ J$.

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13 · Part 1 Core Melting, boiling and evaporation [1]

Answer: A

Explanation

At standard atmospheric pressure, pure ice melts at 0 °C and pure water boils at 100 °C, so A is right. B gives the kelvin values with a °C unit. C confuses the melting point with absolute zero. D uses 4 °C, which is not the melting point of ice.

14 · Part 1 Core Melting, boiling and evaporation [1]

Answer: C

Explanation

In evaporation the more energetic (faster) particles escape from the surface, so the average kinetic energy of the particles left behind falls and the liquid cools; C is right. A reverses which particles escape. B describes boiling, not evaporation. D is wrong because particles do not change size.

15 · Part 1 Supplement Melting, boiling and evaporation [1]

Answer: B

Explanation

Boiling happens throughout the liquid at the boiling point, while evaporation happens only at the surface and at any temperature, so B is right. A and C put evaporation throughout the liquid. D wrongly limits evaporation to the boiling point.

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16 · Part 1 Core States of matter [4]

Mark scheme

(a) **[1]**

- melting (1)

(b) **[1]**

- boiling (1)

ALLOW evaporation / evaporating / vaporisation

(c) **[1]**

- solidification (1)

ALLOW solidifying / freezing

(d) **[1]**

- has a fixed volume (1)

ALLOW cannot be compressed easily / is (much) denser

REJECT has a fixed shape

Worked solution

Solid to liquid is melting. Liquid to gas is boiling (or evaporation, if it happens only at the surface below the boiling point). Liquid to solid is solidification, also called freezing. A liquid has a fixed volume and is hard to compress, but it has no fixed shape. A gas has neither a fixed shape nor a fixed volume and can be compressed easily.

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17 · Part 1 Core Particle model [6]

Mark scheme

(a) **[2]**

- particles in a regular (fixed) pattern / arrangement (1)
- particles vibrate about fixed positions (1)

ALLOW closely packed in a lattice for the first mark

REJECT particles do not move at all

(b) **[2]**

- particles in steam are (very) far apart, compared with close together (touching) in ice (1)
- particles in steam move randomly / freely in all directions (at high speed), instead of only vibrating (1)

(c) **[2]**

- the number of particles (molecules) stays the same (1)
- the particles are much further apart in steam / there is much more space between them (1)

REJECT the particles get bigger

Worked solution

In ice, the particles are packed closely in a regular pattern and can only vibrate about fixed positions. In steam, the particles are far apart, randomly arranged and move quickly in all directions. When water boils, no particles are made or destroyed, and the particles do not change size. The same number of particles simply spread out with much larger gaps between them, so the gas takes up a much larger volume.

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18 · Part 1 Core Particle model [5]

Mark scheme

(a) **[1]**

- random / jerky / zigzag motion, constantly changing direction (1)

(b) **[3]**

- water molecules move randomly (and quickly) (1)
- water molecules collide with / hit / bombard the fat droplets (1)
- at any instant the collisions on different sides of a droplet are unequal / uneven, so the droplet is pushed in changing directions (1)

REJECT fat droplets moved by convection currents

(c) **[1]**

- the kinetic particle model (of matter) (1)

ALLOW liquids / water are made of (small) moving particles

Worked solution

The fat droplets jiggle about randomly, changing direction all the time. The water around them is made of molecules that are far too small to see but are always moving randomly at high speed. These molecules keep hitting the fat droplets. At any moment, more molecules (or harder hits) may push on one side of a droplet than the other, so the droplet moves a little in that direction; a moment later the imbalance is different. This random motion (Brownian motion) is evidence that matter is made of tiny moving particles.

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19 · Part 1 Supplement Particle model [6]

Mark scheme

(a) **[2]**

- in steel, strong forces (of attraction) hold the particles in fixed positions so they can only vibrate (1)
- in water, the forces are weaker / particles are not held in fixed positions, so the particles can move past / slide over each other (1)

(b) **[2]**

- particles in air (a gas) are far apart, with large spaces between them, so they can be pushed closer together (1)
- particles in water (a liquid) are close together / touching, with very little space between them (1)

(c) **[2]**

- particles collide with the walls and each collision exerts a force on the wall (1)
- pressure is the total force (from all the collisions) per unit area (of wall) (1)

Worked solution

(a) The properties of a substance depend on the forces between its particles and how the particles move. In a solid such as steel, strong attractive forces hold each particle in a fixed position, so the bar keeps its shape. In a liquid the particles are still close together but they are not held in fixed positions, so they slide past each other and the liquid flows into the shape of its container.

(b) In a gas the particles are far apart, so there is plenty of empty space and the gas can be squashed. In a liquid the particles are already almost touching, so the liquid can hardly be compressed.

(c) Air particles move randomly and keep hitting the walls of the syringe. Each collision pushes on the wall with a small force. The very large number of collisions gives a total force on the wall, and the force acting on each unit area of wall is the pressure.

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20 · Part 1 Core Particle model [6]

Mark scheme

(a) **[1]**

- the particles move faster / have more kinetic energy (1)

(b) **[2]**

- particles collide with the walls more often / more frequently (1)
- particles hit the walls with greater force / harder (1)

REJECT the particles get bigger / expand

(c) **[2]**

- absolute zero (1)
- −273 °C (1)

(d) **[1]**

- the particles have the least (minimum) kinetic energy (1)

ALLOW the lowest possible kinetic energy

Worked solution

When the gas is heated, its particles gain kinetic energy and move faster. Because the volume is fixed, the faster particles reach the walls more often and each collision is harder, so the force on the walls, and therefore the pressure, increases. If a gas is cooled, its particles slow down. There is a lowest possible temperature, called absolute zero, which is −273 °C (0 K). At absolute zero the particles have the least kinetic energy they can have.

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21 · Part 1 Supplement Gases and the absolute scale of temperature [7]

Mark scheme

(a) **[3]**

- p1V1 = p2V2 or pV = constant (1)
- $1.0 \times 10^5 \times 240 = p2 \times 80$ (1)
- $p2 = 3.0 \times 10^5$ Pa (1)

ALLOW 300 000 Pa / 300 kPa

ALLOW full marks for correct answer with no working

(b) **[2]**

- the same number of particles is in a smaller volume / there are more particles per unit volume (1)
- particles collide with the walls more often / more frequently (per unit area) (1)

REJECT particles move faster / particles hit harder (temperature is constant)

(c) **[1]**

- 120 (1)

(d) **[1]**

- a curve in which the pressure decreases as the volume increases (not a straight line) (1)

ALLOW a curve showing pressure inversely proportional to volume

REJECT a straight line

Worked solution

(a) For a fixed mass of gas at constant temperature, pV = constant, so p1V1 = p2V2.

$1.0 \times 10^5 \times 240 = p2 \times 80$

$p2 = (1.0 \times 10^5 \times 240) / 80 = 3.0 \times 10^5$ Pa.

Check: the volume becomes one third as large (240 / 80 = 3), so the pressure becomes three times as large: $3 \times 1.0 \times 10^5 = 3.0 \times 10^5$ Pa.

(b) No air escapes, so the same number of particles is squeezed into a smaller space. The temperature is the same, so the particles move at the same speed, but they hit each part of the wall more often. More collisions per second on each unit area means a larger pressure.

(c) For every row, pressure × volume is the same: 240 × 100 = 160 × 150 = 100 × 240 = 24 000. So X × 200 = 24 000, and X = 120 kPa.

(d) As the volume increases the pressure decreases, but not in a straight line: the graph is a curve that gets less steep as the volume increases.

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22 · Part 1 Core Gases and the absolute scale of temperature [5]

Mark scheme

(a) **[1]**

- the pressure decreases (1)

(b) **[2]**

- the particles spread out over a larger volume / are further apart (on average) (1)
- the particles collide with the walls less often / less frequently (1)

REJECT particles move more slowly (temperature is constant)

(c) **[1]**

- 295 K (1)

(d) **[1]**

- −196 °C (1)

REJECT 196 °C (sign missing)

Worked solution

(a), (b) The same number of air particles now has more space. At the same temperature they move at the same speed, but they reach the walls less often, so there are fewer collisions per second on each part of the wall and the pressure falls.

(c) $T = \theta + 273 = 22 + 273 = 295$ K. Check: $295 - 273 = 22$ °C.

(d) $\theta = T - 273 = 77 - 273 = -196$ °C. Check: $-196 + 273 = 77$ K.

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23 · Part 1 Supplement Thermal expansion of solids, liquids and gases [6]

Mark scheme

(a) **[1]**

- steel, water, air (1)

(b) **[2]**

- in steel the particles are closely packed in fixed positions and held by strong forces, so as they vibrate more they move only slightly further apart (1)
- in air the particles are far apart with (almost) no forces between them, so when they move faster they spread out much more (1)

(c) **[2]**

- water expands more than glass (when heated) (1)

ALLOW water expands (when heated) for the first mark

- there is no space for the water to expand into, so it pushes on the bottle with a large force / pressure (1)

(d) **[1]**

- the cables contract in cold weather, and if they were tight they could snap / pull on the pylons (1)

Worked solution

(a) For the same temperature rise, gases expand most, liquids less and solids least: steel, then water, then air.

(b) In a solid, strong forces hold the particles close together in fixed positions; heating makes them vibrate more strongly, so the average separation increases only a little. In a gas, the particles are already far apart and the forces between them are very small, so when they speed up nothing holds them together and the gas expands a great deal.

(c) Both the glass and the water expand as they warm up, but the water expands much more. The bottle is full and sealed, so the water has nowhere to go. It pushes outwards on the glass with a very large force, which can crack it.

(d) On a cold winter night the cables will be colder than on the day they were put up, so they will contract and become shorter. If they were already tight, the contraction could break them or damage the pylons, so slack is left to allow for this.

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24 · Part 1 Supplement Specific heat capacity [6]

Mark scheme

(a) **[1]**

- it increases (1)

(b) **[1]**

- the average kinetic energy of the particles increases (1)

> **ALLOW** the particles vibrate faster / with more energy

(c) **[2]**

- $\Delta\theta = \Delta E / (mc)$ or $84\,000 = 2.0 \times 4200 \times \Delta\theta$ (1)
- 10 °C (1)

(d) **[1]**

- 50 °C (1)

> **ECF** from method in (c)

(e) **[1]**

- sand has a lower specific heat capacity, so the same energy gives it a (five times) larger temperature rise (1)

> **ALLOW** answer in terms of the calculated temperature rises, e.g. "the sand's temperature rises by 50 °C but the water's only by 10 °C"

Worked solution

(a), (b) A rise in temperature means the internal energy of the sand increases; the particles of the sand have a higher average kinetic energy.

(c) $c = \Delta E / (m\Delta\theta)$, so $\Delta\theta = \Delta E / (mc) = 84\,000 / (2.0 \times 4200) = 84\,000 / 8400 = 10$ °C.

(d) $\Delta\theta = 84\,000 / (2.0 \times 840) = 84\,000 / 1680 = 50$ °C.

Check: $2.0 \times 4200 \times 10 = 84\,000$ J and $2.0 \times 840 \times 50 = 84\,000$ J.

(e) Each kilogram of sand needs much less energy than water to warm by 1 °C. With the same energy and the same mass, the sand warms by 50 °C but the water by only 10 °C, so the sand becomes much hotter.

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25 · Part 1 Supplement Specific heat capacity [7]

Mark scheme

(a) **[4]**

- measure the mass of the oil with the balance (e.g. mass of beaker with oil minus mass of empty beaker) (1)
- measure the starting temperature of the oil with the thermometer (1)
- heat the oil with the immersion heater, stirring, then switch off and record the highest / final temperature (1)
- record the energy supplied from the joulemeter (difference between the readings at the start and end) (1)

ALLOW wrap the beaker in insulation / use a lid to reduce energy loss, in place of any one of the first three marking points

(b) **[2]**

- c = 16 500 / (0.50 × 15.0) (1)
- 2200 J / (kg °C) (1)

ALLOW 2.2×10^3 J / (kg °C)

ALLOW full marks for correct answer with no working

(c) **[1]**

- energy is lost to the surroundings, so not all the energy measured heats the oil (1)

ALLOW some energy heats the beaker / thermometer / heater

REJECT "human error" or "inaccurate readings" with no explanation

Worked solution

(a) Find the mass of oil: weigh the empty beaker, add the oil and weigh again; the difference is the mass of oil. Insulate the beaker. Put the heater and thermometer in the oil and record the starting temperature and the joulemeter reading. Switch on the heater and stir the oil so it is heated evenly. After the temperature has risen by about 10–20 °C, switch off, keep stirring and record the highest temperature and the new joulemeter reading. Then $c = \Delta E / (m\Delta\theta)$.

(b) $\Delta\theta = 34.0 - 19.0 = 15.0$ °C.

$c = \Delta E / (m\Delta\theta) = 16\,500 / (0.50 \times 15.0) = 16\,500 / 7.5 = 2200$ J / (kg °C).

Check: $0.50 \times 2200 \times 15.0 = 16\,500$ J.

(c) Some of the energy from the heater escapes to the surroundings or warms the beaker, so the measured ΔE is bigger than the energy that actually heated the oil. Dividing a too-large ΔE by the same mΔθ gives a value of c that is too high.

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26 · Part 1 Core Melting, boiling and evaporation [6]

Mark scheme

(a) **[1]**

- 54 °C (1)

(b) **[1]**

- solid and liquid (both present) (1)

ALLOW the substance is melting

REJECT solid only or liquid only

(c) **[2]**

- the energy is used to overcome / break the forces of attraction between particles (so the particles can move apart / out of fixed positions) (1)
- the kinetic energy of the particles does not increase, so the temperature stays the same (1)

(d) **[2]**

- the particles lose energy / move more slowly (1)
- the forces of attraction pull the particles into fixed positions (in a regular pattern), where they only vibrate (1)

Worked solution

(a) The temperature stays at 54 °C while the substance changes from solid to liquid, so the melting point is 54 °C.

(b) At 8 minutes the substance is in the middle of the flat part of the data, so it is melting: some is still solid and some is liquid.

(c) While a solid melts, the energy supplied is used to separate the particles by working against the attractive forces that held them in fixed positions. It is not used to make the particles move faster, so their kinetic energy, and therefore the temperature, does not change until all of the solid has melted.

(d) When the liquid cools, its particles lose energy and slow down. The attractive forces between them pull them into fixed positions in a regular arrangement, and from then on they can only vibrate, so the substance becomes a solid.

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27 · Part 1 Core Melting, boiling and evaporation [4]

Mark scheme

(a) **[1]**

- evaporation (1)

(b) **[2]**

- the more energetic / faster particles (molecules) (1)
- at the surface of the water escape into the air (1)

(c) **[1]**

- it decreases (1)

ALLOW the water cools down

Worked solution

The water disappears by evaporation. The water particles have a range of energies. The more energetic ones near the surface are moving fast enough to escape into the air. Because the most energetic particles leave, the water left behind has less energetic particles on average, so its temperature falls and the swimmer feels cold.

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28 · Part 1 Supplement Melting, boiling and evaporation [6]

Mark scheme

(a) **[2]**

- the rate of evaporation increases (1)
- (at a higher temperature) more of the particles have enough energy to escape from the surface (1)

(b) **[2]**

- the rate of evaporation increases (1)
- the surface area is larger, so more particles are at the surface where they can escape (1)

(c) **[2]**

- the rate of evaporation increases (1)
- moving air carries away the water particles that have escaped, so fewer return to the towel (1)

Worked solution

Evaporation happens when energetic particles at the surface of a liquid escape.

(a) In a warmer room the water is warmer, so a larger fraction of its particles have enough energy to escape each second.

(b) Spreading the towel out exposes more wet surface, so more particles are at the surface and able to escape at the same time.

(c) Without air movement, escaped water particles collect near the towel and some fall back into it. Moving air sweeps them away, so fewer return and the overall rate of evaporation is greater.

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29 · Part 1 Supplement Melting, boiling and evaporation [5]

Mark scheme

(a) **[3]**

- the most energetic water particles escape from the outer surface of the pot (1)
- the average kinetic energy of the water left at the surface falls, so its temperature falls (1)
- energy is transferred from the (warmer) water inside the pot, through the clay, to the cooler water at the surface (1)

ALLOW thermal energy flows from the pot and its contents to the evaporating liquid

(b) **[1]**

- moving air removes the escaped water particles, so the rate of evaporation (and so of cooling) increases (1)

(c) **[1]**

- no water reaches the outer surface, so there is no evaporation (to cool the pot) (1)

Worked solution

(a) At the outer surface, the fastest water particles escape, so the water there is left with particles of lower average kinetic energy: it becomes cooler than the water inside. Energy then flows from the warmer water inside the pot, through the clay, to this cooler surface water. As this keeps happening, the water inside the pot loses energy and its temperature falls.

(b) A breeze carries the escaped water particles away, so fewer return and evaporation, and therefore cooling, is faster.

(c) The glaze stops water reaching the outside. With no evaporation from the outer surface, nothing removes energy from the pot, so its contents stay at the temperature of the air.

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30 · Part 1 Supplement Melting, boiling and evaporation [5]

Mark scheme

(a) **[2]**

- boiling happens throughout the liquid but evaporation happens only at the surface (1)
- boiling happens only at the boiling point (a fixed temperature) but evaporation happens at any temperature (1)

> **ALLOW** boiling produces bubbles (of vapour) in the liquid but evaporation does not

> **ALLOW** boiling needs a continuous energy supply but evaporation does not / evaporation cools the liquid

(b) **[2]**

- the steam particles lose energy / slow down when they hit the cold glass (1)
- the attractive forces pull the particles close together, so they form a liquid (1)

(c) **[1]**

- the particles have a range of energies / speeds, and some at the surface have enough energy to escape (1)

Worked solution

(a) Boiling happens all through the liquid, with bubbles of vapour forming inside it, and only at the boiling point. Evaporation happens only at the surface of the liquid and can take place at any temperature.

(b) Steam particles are far apart and moving fast. When they hit the cold window they transfer energy to the glass and slow down. The attractive forces between the slower particles pull them close together, and they form droplets of liquid water: this is condensation.

(c) Even at 18 °C the water particles do not all have the same energy: some move much faster than the average. Any of these more energetic particles that reach the surface can escape into the air, so the puddle slowly evaporates.

1 · Part 2 Core Conduction [1]

Answer: B

Explanation

Brass is a good thermal conductor, so it conducts energy away from the paper quickly and the paper does not get hot enough to scorch; wood is a thermal insulator, so the paper over it heats up and scorches. A reverses the result. C is wrong because the paper, not the brass, faces the flame, and the effect is due to conduction. D is wrong because the paper is wrapped tightly at the same distance all round.

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2 · Part 2 Supplement Conduction [1]

Answer: D

Explanation

In metals, free (delocalised) electrons move through the metal and transfer energy quickly, in addition to lattice vibrations. A is wrong because atoms in all solids vibrate more when heated. B is wrong because atoms in a solid metal stay in fixed positions in the lattice. C is wrong because the mass of the atoms is not the reason metals conduct well.

3 · Part 2 Supplement Conduction [1]

Answer: A

Explanation

Gas particles are far apart, so collisions between them are much less frequent and energy is passed from particle to particle only slowly. B is wrong because gas particles move rapidly and randomly. C is wrong because particle size is not the reason (and gas particles are not larger). D is wrong because gases have a much lower density than solids.

4 · Part 2 Supplement Conduction [1]

Answer: C

Explanation

There is a range of thermal conductivity: many solids (for example glass, brick or concrete) conduct better than insulators such as expanded polystyrene but much less well than metals such as copper. A ignores these in-between materials. B is wrong because non-metal solids conduct by lattice vibrations. D is wrong for the same reason: having no free electrons makes conduction poorer, not impossible.

5 · Part 2 Core Convection [1]

Answer: A

Explanation

When water is heated it expands, so the same mass takes up more volume and its density decreases; the less dense warm water rises and cooler, denser water sinks. B and C are common misconceptions: the particles themselves do not get lighter or expand. D is wrong because it is the warm, less dense fluid that rises, not "heat".

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6 · Part 2 Core Convection [1]

Answer: D

Explanation

Water is a liquid, so it can flow: water heated by the element at the bottom becomes less dense and rises, setting up a convection current. A and C are conduction through solids. B is radiation, because space is almost a vacuum.

7 · Part 2 Core Radiation [1]

Answer: B

Explanation

Thermal radiation is infrared radiation. Ultraviolet, microwaves and X-rays are other regions of the electromagnetic spectrum and are not what is meant by thermal radiation.

8 · Part 2 Core Radiation [1]

Answer: C

Explanation

Conduction and convection both need particles (a medium) to pass on or carry the energy; space is almost a vacuum, so only radiation, which needs no medium, can cross it. A and B are not the reasons. D is wrong because convection happens in liquids and gases, not solids.

9 · Part 2 Core Radiation [1]

Answer: D

Explanation

Dull black surfaces are the best emitters of infrared radiation, so the dull black can emits at the greatest rate. Shiny surfaces and white surfaces are poorer emitters; the shiny silver can is the worst emitter.

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10 · Part 2 Supplement Radiation [1]

Answer: A

Explanation

An object stays at a constant temperature only when it transfers energy away at the same rate as it receives energy. B and C are wrong because all objects emit infrared radiation and objects can receive energy while at constant temperature. D would make the temperature rise.

11 · Part 2 Supplement Radiation [1]

Answer: B

Explanation

The rock receives energy faster than it transfers energy away, so its temperature rises. As its temperature rises it transfers energy away (including by emitting infrared) at a greater rate, until the rate away equals the 40 W it receives; then its temperature stays constant. A is wrong because transferring some energy away is not enough: the rates must be equal. C is wrong because the rate of transfer away increases as the rock gets hotter, so the rise slows and stops. D is wrong because it receives more energy per second than it loses.

12 · Part 2 Supplement Radiation [1]

Answer: C

Explanation

If the atmosphere absorbs more of the infrared emitted by the surface, less energy escapes into space, so the Earth receives energy faster than it loses it and its temperature rises. A is wrong: ice and snow reflect more incoming radiation, which causes cooling. B is wrong: less incoming energy causes cooling. D is wrong: losing energy faster with unchanged input causes cooling.

13 · Part 2 Supplement Radiation [1]

Answer: A

Explanation

The intensity of radiation reaching the detector depends on its distance from the face, so a different distance for one face would make the comparison unfair. B, C and D all help to keep the test fair: same detector, nearly the same water temperature for every reading, and the same aim for each face.

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14 · Part 2 Supplement Radiation [1]

Answer: D

Explanation

The rate of emission of infrared radiation increases with surface temperature and with surface area. D has both the larger surface area and the higher temperature. A has both the smaller area and the lower temperature, so it emits at the lowest rate; B and C each have only one of the two larger values.

15 · Part 2 Core Consequences of thermal energy transfer [1]

Answer: B

Explanation

A heater near the floor warms the air next to it; this air expands, becomes less dense and rises, while cooler, denser air sinks to replace it, so a convection current circulates through the whole room. A would leave warm air trapped at the ceiling with little circulation. C is wrong because cool air is denser, not less dense. D is wrong because warm air is less dense, not denser.

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16 · Part 2 Core Conduction [5]

Mark scheme

(a) **[2]**

- any two of: same length of rod (1); same diameter / thickness of rod (1); same amount / size of wax blob (1); same distance of the pin from the flame / same part of rod in the flame (1); same flame / same heating (1)

ALLOW same type of pin / same mass of pin

IGNORE "same material"

(b) **[2]**

- copper (1)
- its pin fell off in the shortest time / thermal energy reached the far end of the copper rod fastest (1)
- the second mark can only be scored if copper is chosen

(c) **[1]**

- (thermal) insulator (1)

ALLOW poor conductor

Worked solution

(a) In a fair test only the material of the rod should change. Anything else that affects how quickly the wax at the far end melts must be kept the same: the length and thickness of each rod, the size of the wax blob, and how the rods are heated (the same flame, with each rod the same distance into it).

(b) The best conductor transfers thermal energy to the far end fastest, so the wax melts first. Copper's pin fell off first, so copper is the best conductor.

(c) A material that is a bad thermal conductor, like glass here, is called a thermal insulator.

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17 · Part 2 Supplement Conduction [6]

Mark scheme

(a) **[2]**

- particles / atoms at the hot end vibrate more (1)
- these vibrations are passed on to neighbouring particles (through the lattice) along the rod (1)

ALLOW atoms gain kinetic energy and collide with / pass energy to neighbouring atoms

(b) **[2]**

- aluminium (is a metal so it) has free / delocalised electrons (1)
- the free electrons move (quickly) through the metal / travel long distances and transfer energy to particles further along the rod (1)

ALLOW aluminium conducts by both lattice vibrations and free electrons, glass only by lattice vibrations

(c) **[2]**

- the statement is incorrect: glass does conduct thermal energy (by lattice vibrations), but only slowly / much less well than metals (1)
- glass conducts better than good thermal insulators (such as air or expanded polystyrene) / glass is between good conductors and insulators (1)

REJECT glass is a perfect insulator

Worked solution

(a) Conduction in all solids happens through lattice vibrations. The atoms at the hot end gain energy and vibrate more strongly. Because the atoms are bonded to their neighbours, these vibrations are passed along from atom to atom, so energy gradually moves along the rod.

(b) Aluminium is a metal, so as well as lattice vibrations it has free (delocalised) electrons. These electrons gain energy at the hot end and move quickly through the metal, carrying energy far along the rod and passing it to atoms by collisions. This is much faster than passing vibrations from atom to atom, so aluminium is a much better conductor.

(c) Being a bad conductor is not the same as not conducting at all. Glass does conduct thermal energy, just slowly. In fact many solids such as glass conduct better than thermal insulators like air or polystyrene foam, but much less well than metals.

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18 · Part 2 Supplement Conduction [5]

Mark scheme

(a) **[2]**

- particles in air (a gas) are far apart (1)
- so they collide (with each other) less often / energy is passed from particle to particle only slowly (1)

(b)(i) **[1]**

- particles in water are (much) closer together (than in air), so they collide more often / pass energy on more easily (1)

(b)(ii) **[2]**

- metals have free / delocalised electrons that transfer energy quickly; water has no free electrons (1)
- water particles are not held in a (regular, fixed) lattice, so energy is passed on only by (random) collisions between particles / not by lattice vibrations (1)

ALLOW metal atoms are closely packed in a lattice so vibrations pass on efficiently

Worked solution

(a) Conduction passes energy from particle to particle. In a gas such as air the particles are very far apart, so they collide with each other only rarely and energy is passed on very slowly. That is why trapped air is a good insulator.

(b)(i) In a liquid the particles are much closer together than in a gas, so they collide more often and energy can be passed on more easily than in air. Replacing the air with water makes the sleeping bag a better conductor, so the person loses energy faster.

(b)(ii) Metals conduct well because they have free electrons that carry energy quickly through the metal, and their atoms are held closely in a lattice so vibrations are passed on efficiently. Water has no free electrons and its particles are not held in a fixed lattice; energy can only be passed on by random collisions between moving particles, which is slow. So water is still a bad conductor compared with a metal.

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19 · Part 2 Core Convection [6]

Mark scheme

(a) **[2]**

- the colour rises (straight) up from the crystal / above the flame (1)
- it then moves across the top (to the right-hand side) and down the other side (and back along the bottom), so it circulates (1)

(b) **[3]**

- the water above the flame is heated and expands (1)
- so its density decreases and it rises (1)
- cooler, denser water sinks (on the other side) to take its place, forming a convection current (1)

REJECT particles expand / particles become less dense (for the second mark)

(c) **[1]**

- particles in a solid cannot move from place to place / the solid cannot flow (1)

ALLOW particles only vibrate about fixed positions

Worked solution

(a) The purple colour shows where the water near the crystal goes. It rises straight up above the flame, spreads across the top of the beaker, moves down the cooler right-hand side and back across the bottom: a loop.

(b) The flame heats the water at the bottom on the left. This water expands, so the same mass takes up more volume and its density decreases. The less dense water rises. Cooler water elsewhere is denser, so it sinks and flows in to replace the rising water. This continuous circulation is a convection current. Remember: the particles do not get bigger; they move further apart.

(c) Convection needs the warmer part of the material to move bodily from one place to another. In a solid the particles are held in fixed positions and can only vibrate, so the solid cannot flow and convection cannot happen.

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20 · Part 2 Core Convection [5]

Mark scheme

(a) **[1]**

- convection (1)

(b) **[1]**

- on top of the cans (1)

(c) **[3]**

- air next to the ice pack is cooled and contracts / becomes denser (1)
- the cooler, denser air sinks down past the cans (1)
- warmer, less dense air rises to the ice pack, is cooled and sinks, so a convection current circulates past all the cans (1)

> **ALLOW** if the ice pack were underneath, the cold dense air would stay at the bottom and the top cans would not be cooled (as an alternative to the third point)

Worked solution

(a) The air inside the box can move, so the main method of thermal energy transfer through it is convection.

(b) The ice pack should go on top of the cans.

(c) The ice pack cools the air next to it. Cold air contracts, so its density increases and it sinks down through the gaps between the cans, gaining energy from each can it passes, so the cans cool down. Warmer, less dense air near the bottom rises up towards the ice pack, where it is cooled and sinks again. This convection current keeps circulating, so every can is cooled. If the ice pack were at the bottom, the cold, dense air would just stay at the bottom and the cans at the top would stay warm.

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21 · Part 2 Core Radiation [6]

Mark scheme

(a) **[2]**

- the student is not correct (1)
- all objects emit thermal (infrared) radiation, including cool / cold objects (1)

ALLOW hotter objects emit more, but every object emits some

- the first mark may only be awarded with a correct explanation

(b)(i) **[2]**

- conduction and convection need a medium / particles (to pass on or carry the energy) (1)
- a vacuum contains no particles / there is no medium in a vacuum (1)

(b)(ii) **[2]**

- by (thermal / infrared) radiation (1)
- radiation does not need a medium / can travel through a vacuum (1)

Worked solution

(a) The student is wrong. Every object emits thermal radiation, which is infrared radiation, whatever its temperature. Very hot objects like the Sun and fires simply emit a lot of it, so we notice it more.

(b)(i) Conduction passes energy from particle to particle, and convection needs a fluid that can flow and carry energy. Both need a medium made of particles. A vacuum has (almost) no particles, so neither conduction nor convection can transfer energy across it.

(b)(ii) Thermal energy can still cross the gap by infrared radiation, because radiation does not need a medium. This is the same way that energy from the Sun crosses space to reach the Earth.

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22 · Part 2 Core Radiation [6]

Mark scheme

(a) **[2]**

- dull black surfaces are good absorbers of infrared radiation (1)
- so more energy (from the Sun) is absorbed / transferred to the water, which gets hotter (1)

(b) **[2]**

- white surfaces are good reflectors / reflect most of the (infrared) radiation from the Sun (1)
- so they absorb less (infrared) radiation / less energy enters the house (1)

ALLOW white is a poor absorber (for either mark, but not both)

(c) **[2]**

- a shiny surface is a poor emitter of infrared radiation, so little energy is radiated from the blanket (1)
- a shiny surface reflects (infrared) radiation emitted by the body back towards the body (1)

ALLOW shiny surface is a good reflector (of infrared) for the second mark

Worked solution

(a) Dull black surfaces absorb infrared radiation well. Painting the pipes dull black means they absorb more of the radiation from the Sun, so more energy passes into the water and it gets hotter.

(b) White surfaces reflect most of the radiation that falls on them and absorb only a little. Less energy from the Sun is absorbed by the walls and roof, so less energy passes into the house and the inside stays cooler.

(c) Shiny surfaces are poor emitters and good reflectors of infrared. The shiny side facing the body reflects much of the infrared emitted by the runner's body back to the body, and the shiny surface itself emits very little infrared to the cold surroundings. Both reduce the energy lost.

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23 · Part 2 Core Radiation [6]

Mark scheme

(a) **[2]**

- X: 19 °C (1)
- Y: 9 °C (1)

(b) **[2]**

- X (is dull black) (1)
- its temperature rose more / faster (19 °C compared with 9 °C), so it absorbed infrared radiation at a greater rate; dull black is the better absorber (1)

ECF from (a)

(c) **[2]**

- any two of: volume / mass of water (1); starting temperature of the water (1); size / shape / material of the cans (1); time between readings / total time (1); same heater / same heater setting (1); whether or not lids are used (1)

IGNORE distance from the heater (given in the question); "same temperature" without saying starting temperature

Worked solution

(a) Temperature rise = final temperature − starting temperature. For X: 37 − 18 = 19 °C. For Y: 27 − 18 = 9 °C.

(b) The water in X warmed up by much more than the water in Y in the same time, so can X absorbed infrared radiation from the heater at a greater rate. Dull black surfaces are better absorbers of infrared than shiny silver surfaces (which reflect most of it), so X is the dull black can.

(c) Only the surface should be different. Anything else that affects how fast the water warms must be the same, for example the volume (mass) of water in each can and its starting temperature, and the size of the cans.

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24 · Part 2 Supplement Radiation [8]

Mark scheme

(a) **[2]**

- the temperature (of the plate) rises (1)
- because it receives / absorbs energy at a greater rate than it transfers energy away (60 W is greater than 15 W) (1)

(b) **[2]**

- net rate of energy gain = 60 − 15 = 45 W (1)
- energy = 45 × 120 = 5400 J (1)

ALLOW 5.4×10^3 J; 5.4 kJ

ALLOW 7200 − 1800 = 5400 J for 2 marks

- answer 7200 J (ignoring the energy transferred away) scores 0

(c) **[2]**

- $\Delta\theta = \Delta E / (mc) = 5400 / (0.40 \times 900)$ (1)
- = 15 °C (1)

ECF from (b)

(d) **[2]**

- as the temperature of the plate rises, it transfers energy to the surroundings (emits infrared radiation) at a greater rate (1)
- the temperature stops rising when the rate of transfer away equals the rate at which it receives energy / reaches 60 W (1)

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Worked solution

(a) The plate receives energy at 60 W but only transfers energy away at 15 W, so it is gaining energy overall. Its internal energy increases, so its temperature rises.

(b) The plate gains 60 J every second and loses 15 J every second, so the net gain is 60 − 15 = 45 J every second (45 W). Using $P = \Delta E / t$, the energy gained is $\Delta E = P \times t = 45 \times 120 = 5400$ J.

(c) Rearrange $c = \Delta E / (m\Delta\theta)$ to give $\Delta\theta = \Delta E / (mc) = 5400 / (0.40 \times 900) = 5400 / 360 = 15$ °C. Check: $0.40 \times 900 \times 15 = 5400$ J, as required.

(d) A hotter object transfers energy away faster, for example it emits infrared radiation at a greater rate. So as the plate warms up, the rate of transfer away rises above 15 W and the net gain gets smaller. When the rate of transfer away has risen to 60 W, the plate loses energy as fast as it gains it, and its temperature stays constant.

25 · Part 2 Supplement Radiation [6]

Mark scheme

(a) **[1]**

- the two rates were (approximately) equal (1)

(b) **[3]**

- less (infrared) radiation (emitted by the surface) escapes into space / more is absorbed and re-emitted back towards the surface (1)
- so the Earth now transfers energy away at a lower rate than it receives energy (1)
- so the temperature rises (until the rate of transfer away again equals the rate received) (1)

REJECT the gases block / reflect incoming sunlight as the main cause

(c) **[2]**

- dark sea / land absorbs more (and reflects less) radiation from the Sun than white ice (1)
- so the Earth absorbs energy at a greater rate / receives more energy than it transfers away, so the temperature rises further (1)

Worked solution

(a) A constant temperature means the energy coming in each second is balanced by the energy going out each second: the two rates are equal.

(b) The Earth's surface emits infrared radiation. Gases that absorb infrared trap some of this energy and send some of it back towards the surface, so less energy escapes into space each second. Now the Earth is losing energy more slowly than it receives it from the Sun, so the average temperature rises. It keeps rising until the Earth is hot enough to emit energy as fast as it receives it again.

(c) White ice is a good reflector and a poor absorber of the Sun's radiation. Dark sea and land absorb much more of it. When ice melts, the Earth absorbs energy from the Sun at a greater rate, so the energy balance is pushed further towards warming and the temperature rises even more.

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26 · Part 2 Supplement Radiation [5]

Mark scheme

(a) **[4]**

- fill the cube with (very) hot water (from the kettle) and put the lid on (1)
- place the detector (pointing at each face in turn) at the same distance from each face, measured with the metre rule (1)
- record the reading of the detector for each face (1)
- take the readings quickly / at the same water temperature (1)
- the face giving the highest reading is the best emitter (1)
- max 4

(b) **[1]**

- shiny silver (1)

Worked solution

(a) Fill the cube with very hot water so that all four faces are at the same temperature (the lid helps to keep it hot). Point the infrared detector at one face, with the detector a fixed distance from it, for example 5 cm measured with the metre rule. Record the reading. Turn the cube so that each face in turn is the same distance from the detector, and record each reading. Do this quickly so that the water temperature hardly changes. Since every face is at the same temperature and the same distance from the detector, the only difference is the surface, so the face with the highest reading is the best emitter.

(b) Shiny surfaces and light-coloured surfaces are poor emitters, so the shiny silver face gives the lowest reading. The dull black face should give the highest.

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27 · Part 2 Supplement Radiation [6]

Mark scheme

(a) **[3]**

- the coin on (dull black) plate P falls off first (1)
- dull black is a better absorber of infrared radiation (than a shiny surface, which reflects most of it) (1)
- so plate P's temperature rises faster and the wax on P melts first (1)
- the second and third marks can only be scored if P is chosen

(b) **[1]**

- so that both plates receive the same amount of radiation / energy per second (so only the surface is different) (1)

(c) **[2]**

- copper is a good thermal conductor (and wood is an insulator), so energy absorbed at the front reaches the wax at the back quickly (1)
- a thin plate has a small mass / heats up quickly, so the wax melts in a short time (1)

Worked solution

(a) Plate P's dull black surface absorbs infrared well, while plate Q's shiny surface reflects most of the infrared that falls on it. So P absorbs energy at a greater rate, its temperature rises faster, and the wax behind P melts first. The coin on P falls off first.

(b) The amount of radiation reaching a plate depends on its distance from the heater. With the heater exactly halfway, both plates receive the same radiation, so any difference in result is caused by the surfaces only.

(c) The energy is absorbed at the front surface but the wax is on the back. Copper is a very good thermal conductor, so the energy is conducted to the back quickly; wood is an insulator and would take a very long time. A thin plate also has a small mass, so its temperature rises quickly and the experiment does not take too long.

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28 · Part 2 Supplement Radiation [6]

Mark scheme

(a) **[1]**

- 30 W (1)

(b) **[2]**

- the fins give the heat sink a large surface area (1)
- a larger surface area (at the same temperature) emits radiation at a greater rate (1)

(c) **[3]**

- the temperature (of the heat sink) rises, because it now receives energy faster than it transfers energy away (45 W is more than 30 W) (1)
- as its temperature rises, it transfers energy away / emits radiation at a greater rate (1)
- the temperature becomes constant again, at a higher value, when the rate of transfer away reaches 45 W (1)

Worked solution

(a) The temperature is constant, so the heat sink must transfer energy to its surroundings at the same rate as it receives it from the chip: 30 W.

(b) The rate at which a surface emits infrared depends on its surface area as well as its temperature. Many thin fins give a very large surface area for the volume of metal, so more radiation is emitted per second.

(c) Now 45 W goes in but only 30 W goes out, so the heat sink gains energy and its temperature rises. A hotter surface transfers energy away faster. The temperature keeps rising until the rate of transfer away has increased to 45 W. Then the rates balance again and the temperature stays constant, but at a higher value than before.

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29 · Part 2 Core Consequences of thermal energy transfer [6]

Mark scheme

(a) **[2]**

- copper is a good thermal conductor (1)
- so thermal energy is transferred quickly from the hotplate (through the base) to the water (1)

(b) **[1]**

- plastic is a bad thermal conductor / thermal insulator, so the handle stays cool enough to hold / does not burn the hand (1)

(c) **[2]**

- a shiny surface is a poor emitter of infrared radiation (1)
- so less thermal energy is lost from the sides (by radiation) (1)

(d) **[1]**

- convection (1)

Worked solution

(a) Copper is a good thermal conductor, so energy passes quickly from the hotplate through the base into the water, heating the water quickly.

(b) Plastic is a thermal insulator. Very little energy is conducted along the handle, so it stays cool enough to hold safely.

(c) The hot sides of the pan emit infrared radiation. A shiny surface is a poor emitter, so it emits less infrared than a dull or dark surface would. Less energy is lost from the sides, so the water stays hot.

(d) Water is a liquid and can flow. Water heated at the bottom becomes less dense and rises, while cooler water sinks to the bottom to be heated, so energy spreads through the water mainly by convection.

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30 · Part 2 Supplement Consequences of thermal energy transfer [7]

Mark scheme

(a)(i) **[1]**

- (infrared / thermal) radiation (1)

(a)(ii) **[2]**

- air (and hot gases) heated by the fire expands / becomes less dense (1)
- so it rises up the chimney, carrying thermal energy with it (by convection) (1)

(b) **[4]**

- conduction (through the metal walls of the tubes) from the coolant into the tubes and fins (1)
- (the fins give) a large surface area (in contact with the air) (1)
- air in contact with the fins is heated (by conduction) and carried away by convection / by the moving air from the fan, and replaced by cooler air (1)
- the hot tubes and fins also emit (infrared) radiation (1)

ALLOW convection in the coolant carrying energy to the tube walls (in place of any one point)

Worked solution

(a)(i) The person is not above the fire and is 2 m away through the air, so they are warmed mainly by infrared radiation from the flames and hot embers.

(a)(ii) The fire heats the air and gases around it. Hot air expands, so its density decreases and it rises. The open chimney lets this hot, less dense air rise straight up and out, taking a lot of energy with it by convection, and cooler air from the room flows in to take its place.

(b) Several methods work together. The hot coolant is in contact with the metal tubes, and energy is conducted through the metal walls of the tubes and into the fins, because metal is a good conductor. The many thin fins give a very large surface area in contact with the air. Air touching the hot fins is heated, and the fan (and the movement of the car) carries this warmed air away and brings in cooler air, so energy is removed by convection. The hot metal surfaces also emit some infrared radiation.

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